Homoscedasticity

The variance of errors (residuals) is **constant** across all values of the independent variable.

Simple terms

Errors are **evenly spread**.

Heteroscedasticity

The variance of errors is **NOT constant**.

Errors increase or decrease as input changes

Homoscedasticity is REQUIRED because

- Gives **reliable predictions**
- Ensures **accurate confidence intervals**
- Makes **p-values trustworthy**

What happens if Heteroscedasticity exists?

- Model coefficients still work ✅
- BUT:
  - ❌ Wrong standard errors
  - ❌ Wrong p-values
  - ❌ Misleading conclusions

👉 Very dangerous in:

- Finance models
- Business predictions
- Research analysis

I observed Homoscedasticity best

Condition	Good/Bad	Reason
Homoscedasticity	✅ Best	Stable error variance
Heteroscedasticity	❌ Problem	Unreliable inference